



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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07-Agosto-2026

<https://github.com/jhonys4444/IBM-SpaceX-Capstone>



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies

- Collected SpaceX Falcon 9 launch data from the **SpaceX API** and **Wikipedia web scraping**
- Performed **data wrangling**, feature engineering, and missing value handling
- Conducted **EDA** using Python visualizations and SQL queries
- Built **interactive Folium maps** and a **Plotly Dash dashboard**
- Trained and compared **Logistic Regression, SVM, Decision Tree, and KNN** models

Summary of results

- Launch success rates increased over time
- Launch site, orbit type, payload mass, and booster version influenced landing outcomes
- Interactive maps and dashboards highlighted operational and geographic patterns
- **Decision Tree achieved the best predictive performance** with a cross-validation accuracy of **0.8893**

Introduction

Project background and context

- SpaceX has significantly reduced launch costs through **reusable Falcon 9 first-stage boosters**.
- Predicting whether a booster will land successfully is valuable for **mission planning, cost estimation, and operational efficiency**.
- This project analyzes historical Falcon 9 launch data using data science and machine learning techniques.

Problems to answer

- Which factors are associated with successful booster landings?
- Do **launch site, orbit type, payload mass, and booster version** influence landing success?
- Can a machine learning model accurately predict whether a Falcon 9 booster landing will be successful?

Section 1

Methodology

Methodology



Data Collection

- SpaceX REST API and Wikipedia web scraping



Data Wrangling

- Filter Falcon 9 launches, handle missing values, create landing class labels and encode categorical variables



Exploratory Data Analysis (EDA)

- Python visualizations and SQL queries and aggregations



Interactive Visual Analytics

- Folium geographic maps and Plotly Dash interactive dashboard



Predictive Analysis

- Feature standardization, train/test split, GridSearchCV hyperparameter tuning
- Logistic Regression, SVM, Decision Tree, and KNN evaluation

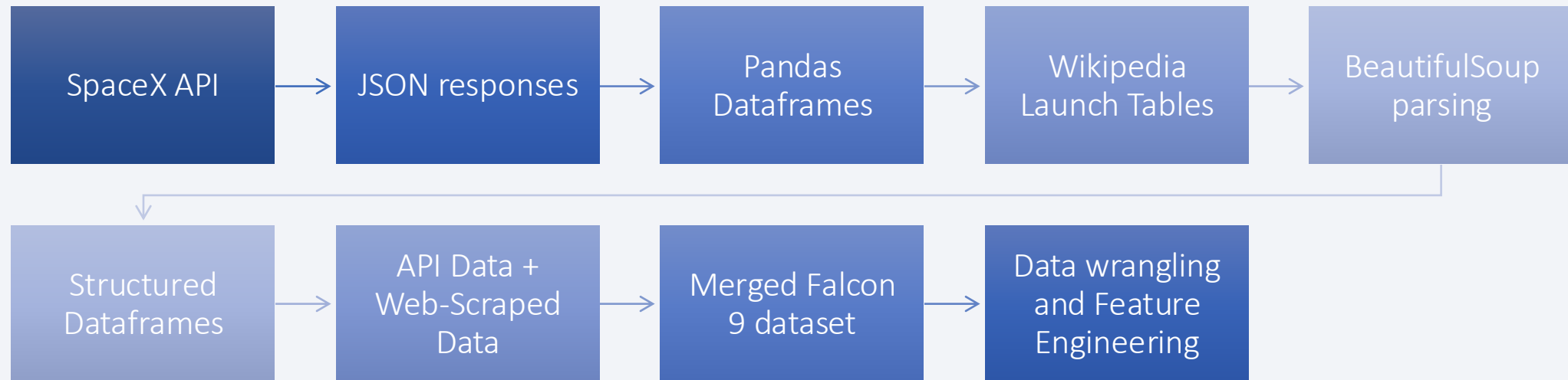
Data Collection

- SpaceX REST API (v4)

- Launch records
- Rocket information
- Payload details
- Launch site coordinates
- Booster core data

- Wikipedia Web Scraping

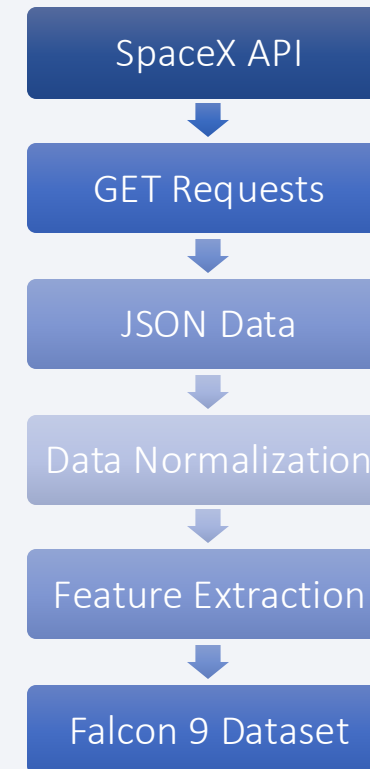
- Falcon 9 and Falcon Heavy launch tables
- Booster versions
- Payload mass
- Orbit type
- Landing outcomes



Data Collection

Key Steps

- SpaceX REST API (v4)
- HTTP GET requests
- JSON responses
- `pandas.json_normalize()`
- Additional endpoints: rockets, payloads, launchpads, cores
- Structured Falcon 9 dataset

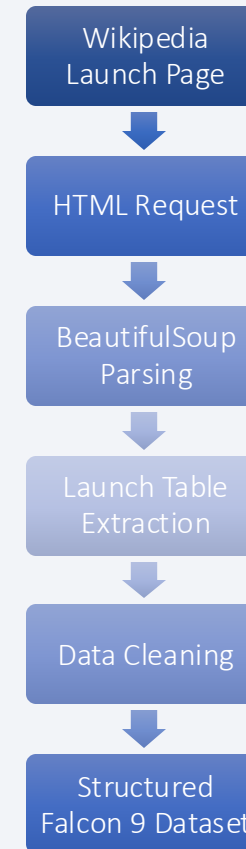


https://github.com/jhonys4444/IBM-SpaceX-Capstone/blob/main/notebooks/01_API-Data-Collection.ipynb

Data Collection

Key Steps

- Wikipedia launch records
- BeautifulSoup parsing
- HTML table extraction
- Data cleaning
- Structured DataFrame



https://github.com/jhonys4444/IBM-SpaceX-Capstone/blob/main/notebooks/02_web_Scraping.ipynb

Data Wrangling

Key Steps

- Merge API and web-scraped data
- Filter Falcon 9 launches
- Remove duplicates
- Handle missing values
- Create landing class labels
- Encode categorical variables



https://github.com/jhonys4444/IBM-SpaceX-Capstone/blob/main/notebooks/03_Data_Wrangling.ipynb

EDA with Data Visualization

Charts Used

- **Scatter plots:** Launch Site vs Flight Number, Payload Mass vs Launch Site
- **Bar charts:** Success rate by Orbit
- **Scatter plots:** Flight Number vs Orbit, Payload Mass vs Orbit
- **Line chart:** Success rate over time (Year)

Purpose

- Identify relationships between **launch site, orbit, payload mass, and landing success**
- Detect **operational trends over time**
- Compare **success rates across launch conditions**

https://github.com/jhonys4444/IBM-SpaceX-Capstone/blob/main/notebooks/05_EDA_Visualization.ipynb

EDA with SQL

- Retrieved **unique launch sites**
- Filtered launches by **site and date**
- Calculated **total and average payload mass**
- Analyzed **landing outcomes** (success vs. failure)
- Identified **booster versions with maximum payload mass**
- Examined **landing outcomes by month and year**
- Ranked **landing outcomes by frequency**

https://github.com/jhonys4444/IBM-SpaceX-Capstone/blob/main/notebooks/04_EDA-SQL.ipynb

Build an Interactive Map with Folium

Map Objects Used

- **Circle markers:** launch site locations
- **Markers:** successful and failed launches
- **Marker clusters:** grouped nearby launch markers
- **Polylines:** distances between launch sites and nearby infrastructure
- **Mouse position tool:** displayed latitude and longitude coordinates

Purpose

- Visualize the **geographic distribution** of launch sites
- Compare **successful vs. failed landings**
- Analyze **proximity to cities, highways, railways, and coastlines**
- Explore launch locations interactively

https://github.com/jhonys4444/IBM-SpaceX-Capstone/blob/main/notebooks/06_Folium_Map.ipynb

Build a Dashboard with Plotly Dash

Dashboard Components

- **Launch Site Dropdown:** filter launches by site
- **Payload Range Slider:** filter launches by payload mass
- **Pie Chart:** compare successful launches across sites
- **Scatter Plot:** analyze payload mass vs. landing outcome by booster version

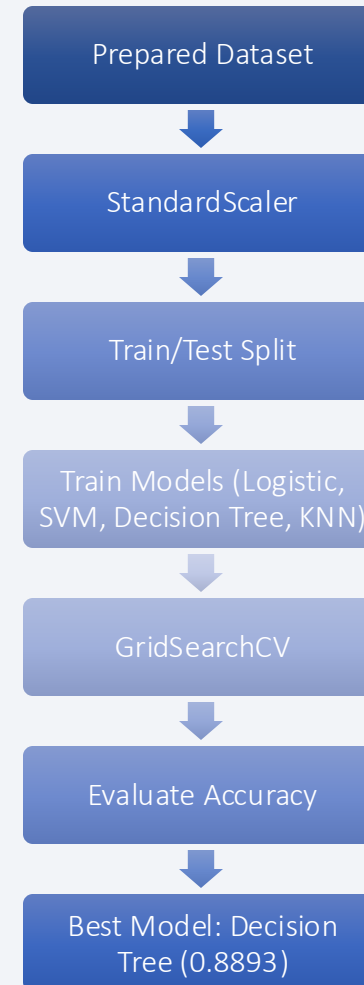
Purpose

- Explore launch performance interactively
- Compare **success rates by launch site**
- Analyze the relationship between **payload mass and landing success**
- Support dynamic filtering and visual exploration of launch data

https://github.com/jhonys4444/IBM-SpaceX-Capstone/blob/main/dash_app/spacex-dash-app.py

Predictive Analysis

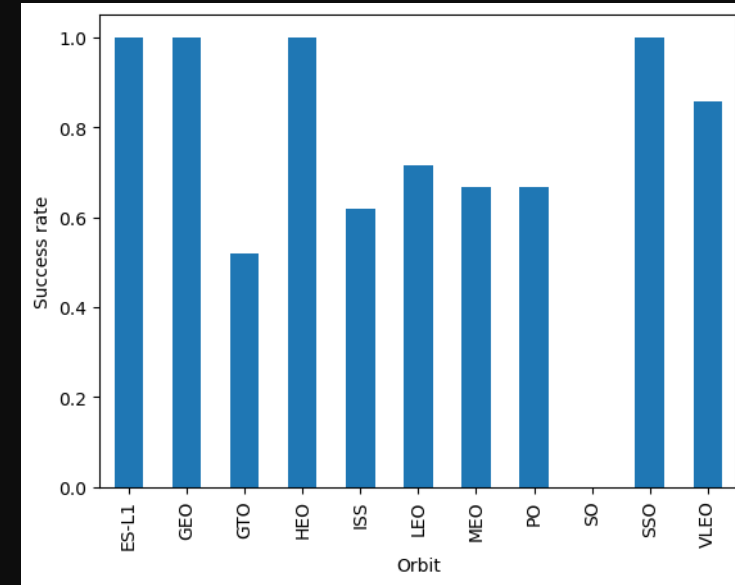
- Feature standardization
- Train/test split
- GridSearchCV hyperparameter tuning
- Cross-validation
- Model evaluation with accuracy and confusion matrix
- Model comparison



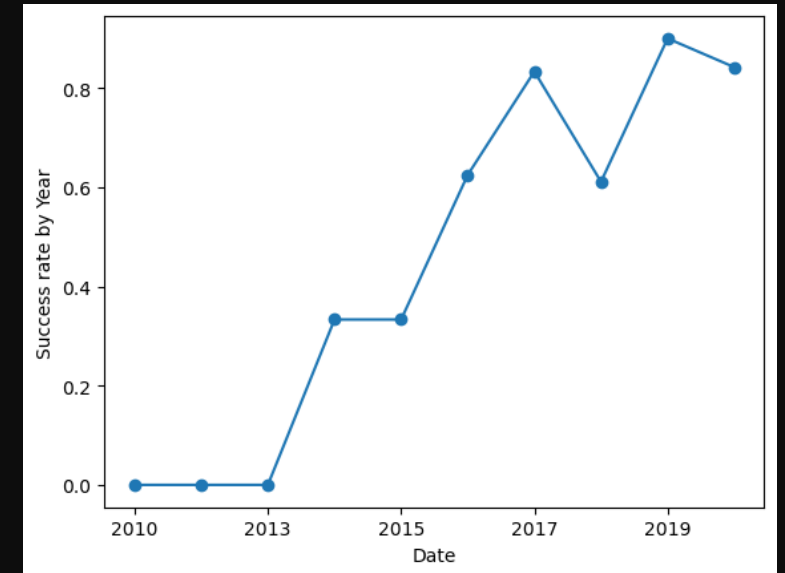
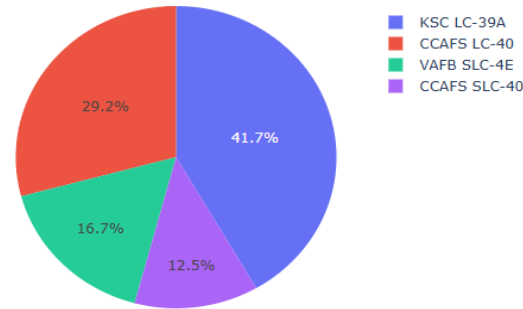
Results

Exploratory Data Analysis Results

- Launch success rates increased over time.
- CCAFS SLC-40 had the highest number of launches.
- Orbit type influenced landing success rates.
- Payload mass showed a relationship with landing outcomes.
- Booster version performance improved across newer Falcon 9 generations.



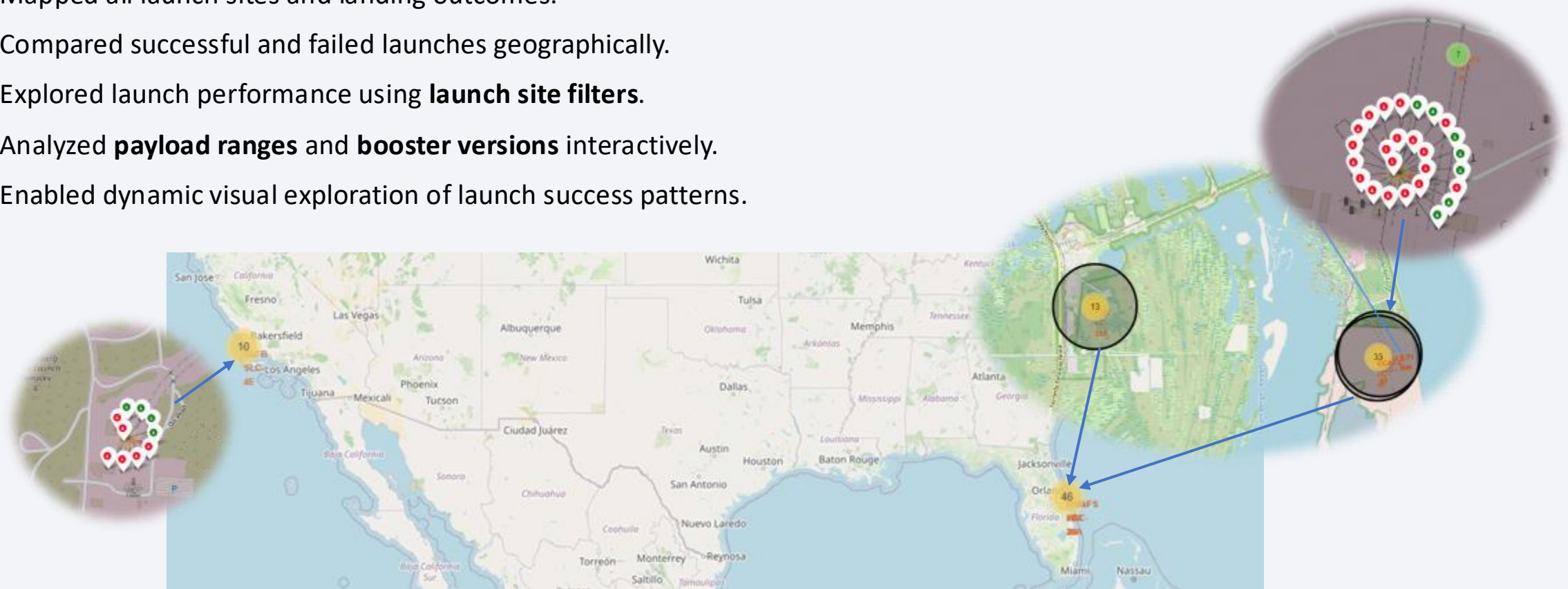
Total Success Launches by Site



Results

Folium and Plotly Dash

- Mapped all launch sites and landing outcomes.
- Compared successful and failed launches geographically.
- Explored launch performance using **launch site filters**.
- Analyzed **payload ranges** and **booster versions** interactively.
- Enabled dynamic visual exploration of launch success patterns.

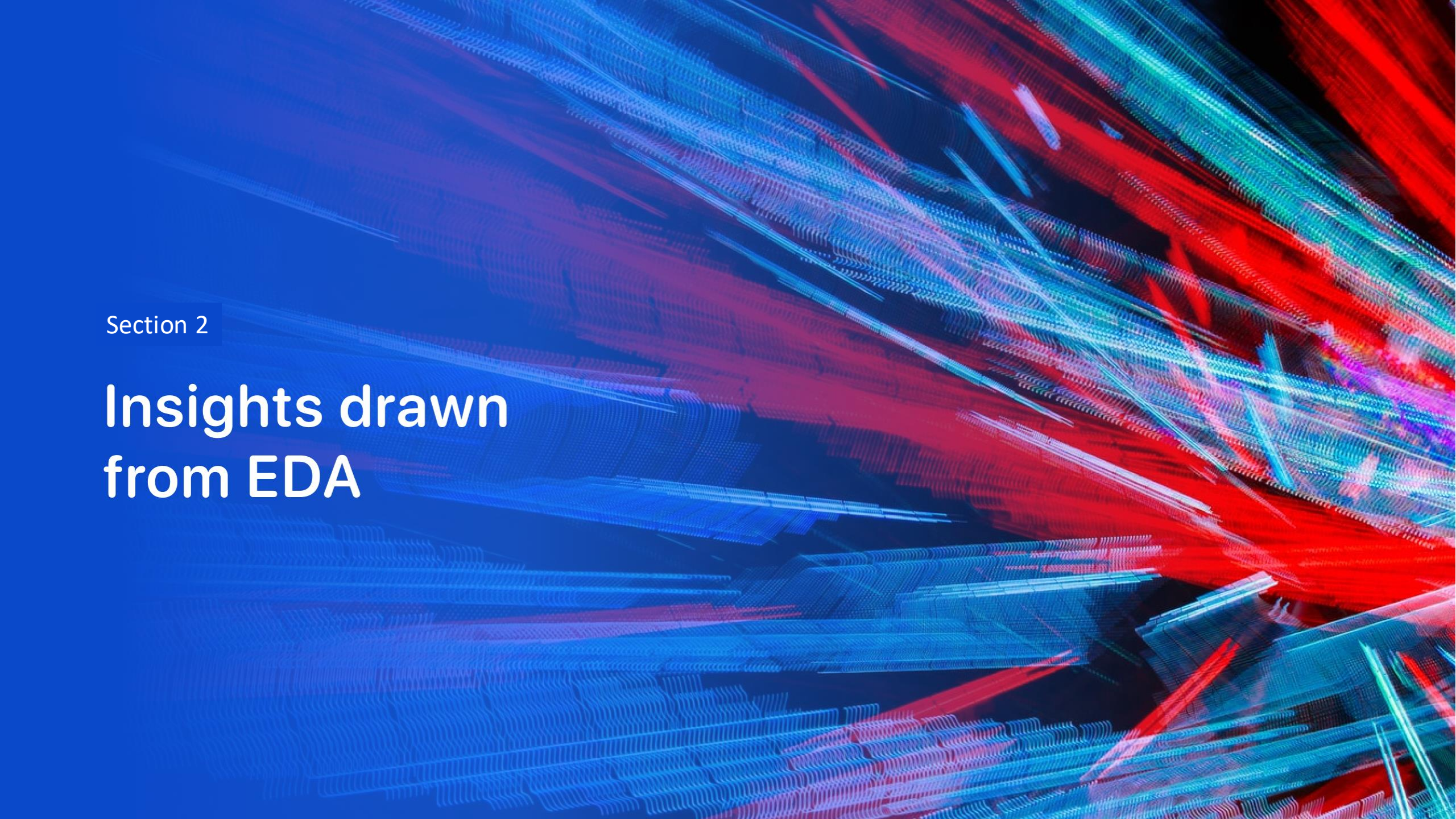


Results

Classification Model Performance

- Features were standardized using **StandardScaler**.
- Models were tuned with **GridSearchCV**.
- Performance was evaluated using **cross-validation, test accuracy, and confusion matrices**.
- **Decision Tree achieved the best overall predictive performance.**

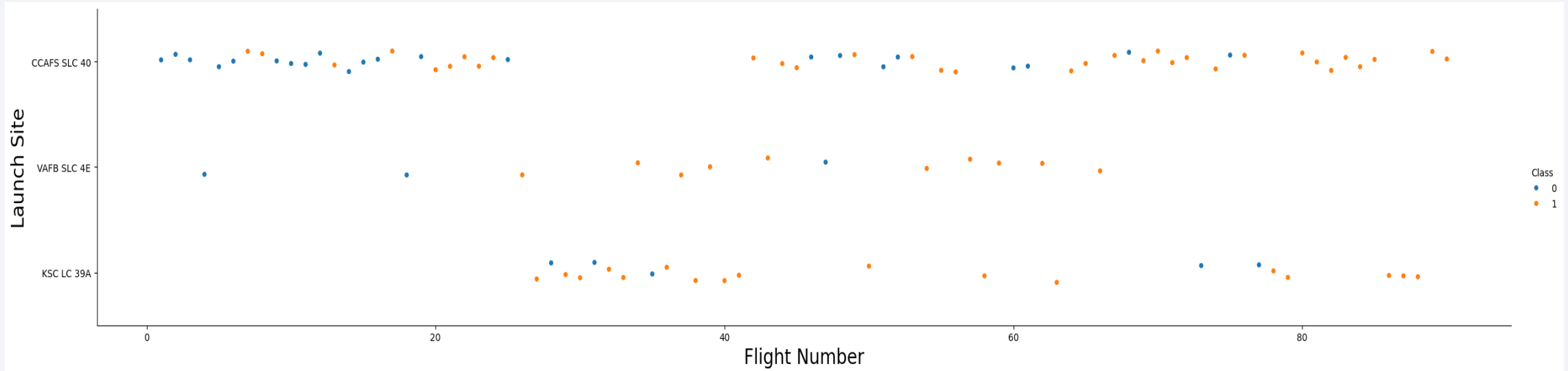
Model	Cross-Validation Accuracy
Logistic Regression	0.8464
SVM	0.8482
Decision Tree	0.8893
KNN	0.8482

The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and teal on the right. These streaks have a textured, almost woven appearance. Overlaid on this pattern is a faint, light blue grid that is more prominent in the lower-left area.

Section 2

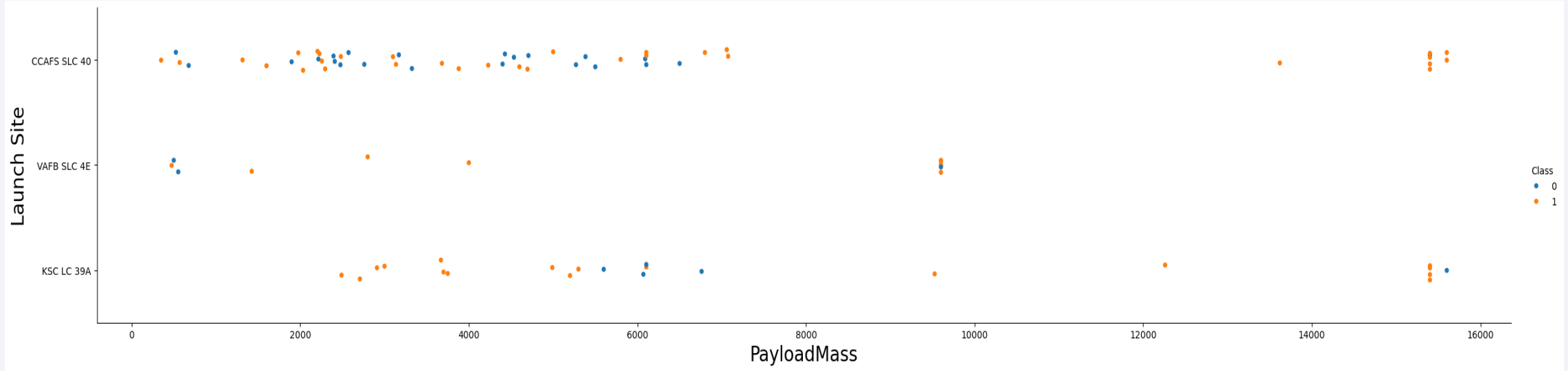
Insights drawn from EDA

Flight Number vs. Launch Site



- **CCAFS SLC-40** handled the largest number of Falcon 9 launches.
- Launch activity increased significantly as **flight numbers progressed over time**.
- **KSC LC-39A** became more active during later missions.
- Later launches showed a **higher concentration of successful landings**, indicating improvements in booster recovery operations.

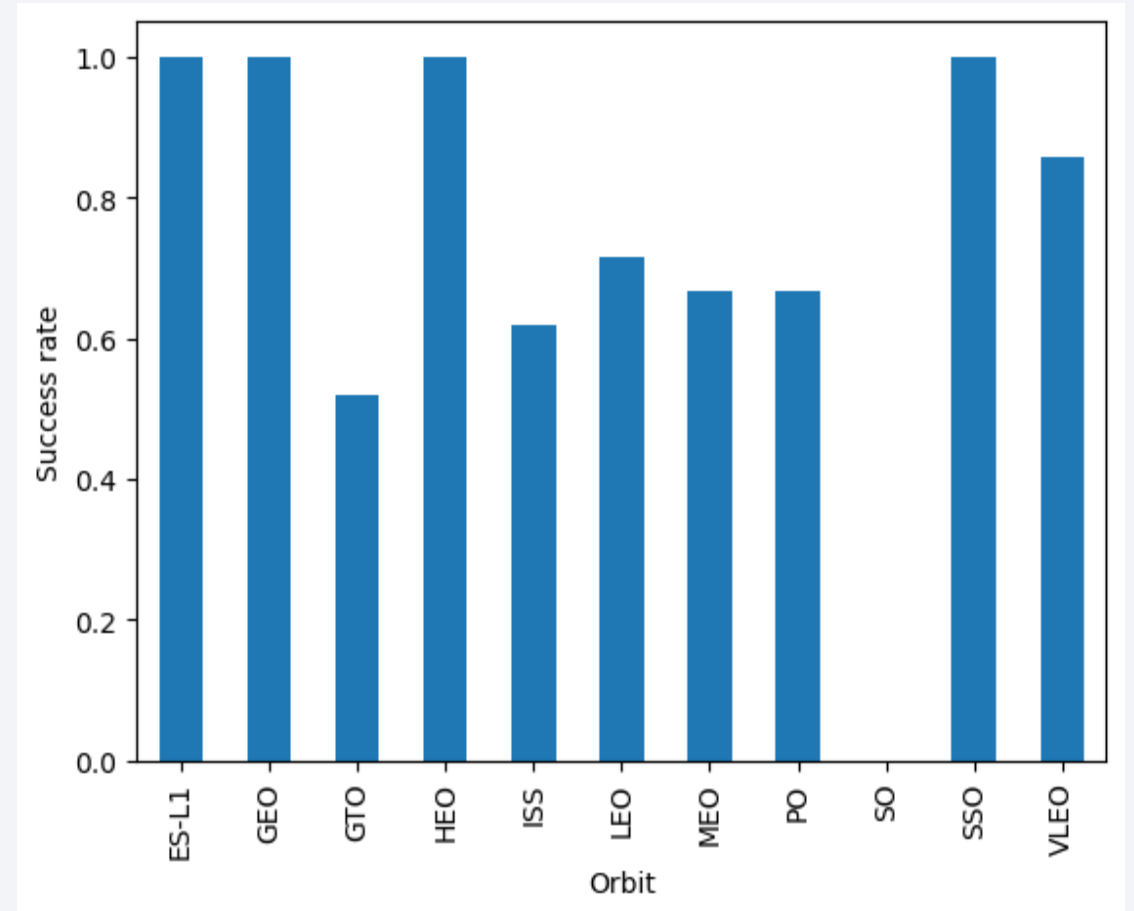
Payload vs. Launch Site



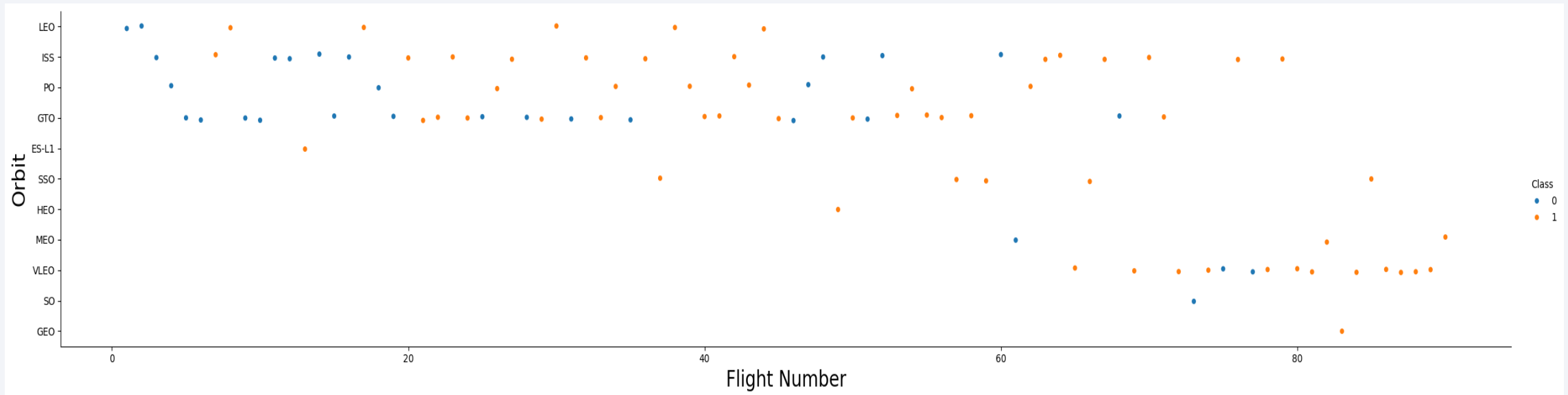
- **CCAFS SLC-40** handled the widest range of payload masses.
- **KSC LC-39A** was associated with several higher-payload missions.
- Higher payload missions generally showed **more successful landings** than lower-payload missions.
- The relationship between **payload mass and landing success** varied across launch sites.

Success Rate vs. Orbit Type

- **ES-L1, GEO, HEO, and SSO** missions achieved the highest observed success rates.
- **GTO and ISS** missions showed consistently high landing success.
- **LEO and PO** missions had comparatively lower success rates.
- Orbit type is an important factor influencing **Falcon 9 booster landing success**.

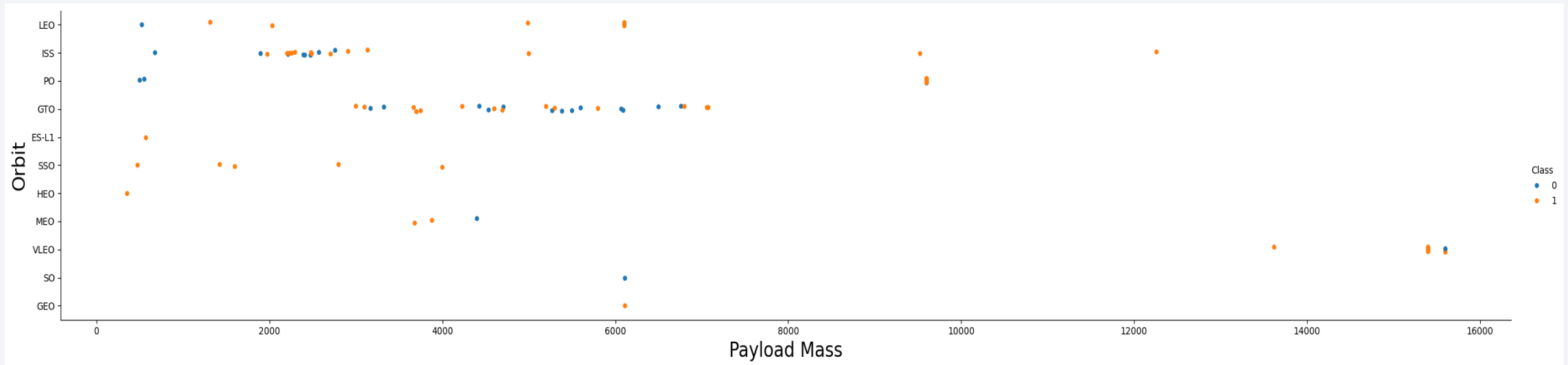


Flight Number vs. Orbit Type



- **GTO** missions became more frequent in later flight numbers.
- **ISS and LEO** missions were concentrated in the earlier Falcon 9 launches.
- Higher flight numbers showed a **greater proportion of successful landings** across multiple orbit types.
- The results indicate that **booster recovery performance improved over time regardless of orbit category**.

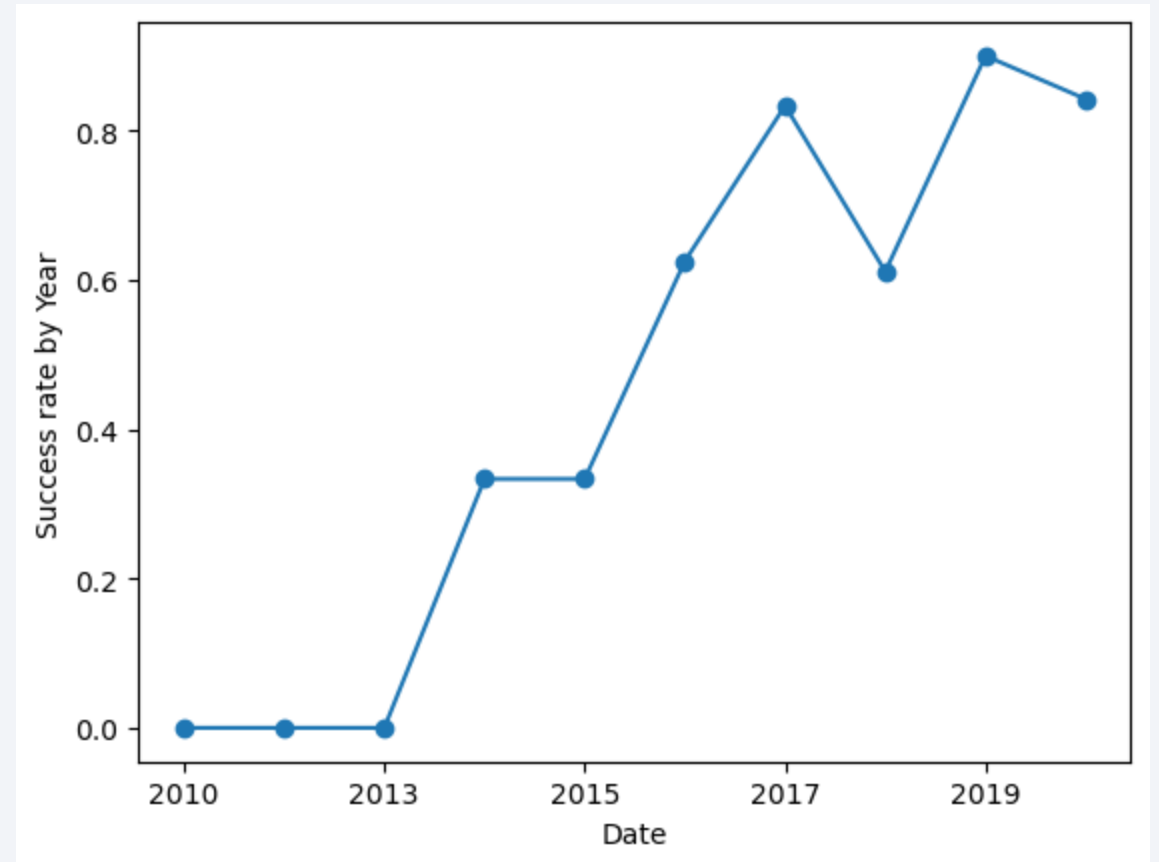
Payload vs. Orbit Type



- **GTO** missions were associated with the highest payload masses.
- **ISS and LEO** missions generally carried lower payloads.
- Successful landings were more common across **medium and high payload ranges**.
- The relationship between **payload mass and landing success** varied by orbit type, indicating that orbit category influences booster recovery performance.

Launch Success Yearly Trend

- The **average launch success rate increased steadily over time.**
- Early Falcon 9 missions had relatively **low landing success rates.**
- A significant improvement occurred after **2015**, with success rates approaching **100%** in later years.
- The trend indicates continuous **advancements in booster recovery technology and operational reliability.**



All Launch Site Names

SQL Query

```
SELECT DISTINCT Launch_Site  
FROM SPACEXTBL;
```

Launch site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

The query retrieves the **unique launch site names** from the SpaceX launch dataset. These four launch sites were used throughout the Falcon 9 missions analyzed in this project and served as key variables for comparing launch activity and landing success rates.

Launch Site Names Begin with 'CCA'

SQL Query

```
SELECT *  
FROM SPACEXTBL  
WHERE Launch_Site LIKE 'CCA%'  
LIMIT 5;
```

Date	Time (UTC)	Booster Version	Launch Site	Payload	Payload Mass (kg)	Orbit	Customer	Mission Outcome	Landing Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

The query uses LIKE 'CCA%' to filter launch sites that begin with **CCA**, returning the first five matching Falcon 9 launch records from **CCAFS LC-40**.

Total Payload Mass

SQL Query

```
SELECT Booster_Version, SUM(PAYLOAD_MASS__KG_) AS  
Total_Payload  
FROM SPACEXTBL  
WHERE Customer = 'NASA (CRS)'  
GROUP BY Booster_Version;
```

Booster Version	Total (kg)	Booster Version	Total (kg)
F9 B4 B1039.2	2647	F9 FT B1025.1	2257
F9 B4 B1039.1	3310	F9 FT B1031.1	2490
F9 B4 B1045.2	2697	F9 FT B1035.1	2708
F9 B5 B1056.2	2268	F9 v1.0 B0006	500
F9 B5 B1058.4	2972	F9 v1.0 B0007	677
F9 B5 B1059.2	1977	F9 v1.1	2296
F9 B5 B1050	2500	F9 v1.1 B1010	2216
F9 B5 B1056.1	2495	F9 v1.1 B1012	2395
F9 FT B1035.2	2205	F9 v1.1 B1015	1898
F9 FT B1021.1	3136	F9 v1.1 B1018	1952

The query calculates the **total payload mass carried by each booster version** for NASA (CRS) missions using SUM(PAYLOAD_MASS__KG_) and GROUP BY Booster_Version. The results show that several **Falcon 9 FT and Block 4/5 boosters** carried the largest cumulative payloads for NASA cargo missions.

Average Payload Mass by F9 v1.1

SQL Query

```
SELECT AVG(PAYLOAD_MASS__KG_) AS Avg_Payload  
FROM SPACEXTBL  
WHERE Booster_Version = 'F9 v1.1';
```

Avg_version_F9_V1_1
2928.4

The query uses `AVG(PAYLOAD_MASS__KG_)` to calculate the **average payload mass carried by the F9 v1.1 booster version**. The result shows that this booster carried an average payload of **2,928.4 kg** across its recorded launches.

First Successful Ground Landing Date

SQL Query

```
SELECT MIN(Date)
FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (ground pad)';
```

MIN(Date)
2015-12-22

The query uses MIN(Date) to identify the **earliest successful ground pad landing** recorded in the dataset. The first successful ground landing occurred on **December 22, 2015**, marking a major milestone in SpaceX reusable booster operations.

Successful Drone Ship Landing with Payload between 4000 and 6000

SQL Query

```
SELECT Booster_Version
FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (drone ship)'
      AND PAYLOAD_MASS__KG_ > 4000
      AND PAYLOAD_MASS__KG_ < 6000;
```

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

The query identifies **Falcon 9 booster versions that successfully landed on a drone ship while carrying payloads between 4,000 and 6,000 kg**. The result shows four Falcon 9 FT boosters that met both the payload and landing-success conditions.

Total Number of Successful and Failure Mission Outcomes

SQL Query

```
SELECT
  CASE
    WHEN Landing_Outcome LIKE 'Success%' THEN 'Success'
    WHEN Landing_Outcome LIKE 'Failure%' THEN 'Failure'
  END AS Outcome_Type,
  COUNT(*) AS Total
FROM SPACEXTBL
WHERE Landing_Outcome LIKE 'Success%'
   OR Landing_Outcome LIKE 'Failure%'
GROUP BY Outcome_Type;
```

Outcome_Type	Total
Failure	10
Success	61

The query groups landing outcomes into **Success** and **Failure** categories using a CASE expression and counts the number of missions in each category. The results show **61 successful landings and 10 failed landings**, indicating a high overall landing success rate in the analyzed SpaceX missions.

Boosters Carried Maximum Payload

SQL Query

```
SELECT Booster_Version
FROM SPACEXTBL
WHERE PAYLOAD_MASS__KG_ = (
    SELECT MAX(PAYLOAD_MASS__KG_)
    FROM SPACEXTBL );
```

Booster Version	Booster Version
F9 B5 B1048.4	F9 B5 B1049.5
F9 B5 B1049.4	F9 B5 B1060.2
F9 B5 B1051.3	F9 B5 B1058.3
F9 B5 B1056.4	F9 B5 B1051.6
F9 B5 B1048.5	F9 B5 B1060.3
F9 B5 B1051.4	F9 B5 B1049.7

The query uses a subquery with `MAX(PAYLOAD_MASS__KG_)` to identify **all booster versions that carried the highest payload mass recorded in the dataset**. The results show that several **Falcon 9 Block 5 boosters** shared the same maximum payload value.

2015 Launch Records

SQL Query

```
SELECT
    SUBSTR(Date,6,2) AS Month,
    Landing_Outcome,
    Booster_Version,
    Launch_Site
FROM SPACEXTBL
WHERE Landing_Outcome = 'Failure (drone ship)'
    AND SUBSTR(Date,1,4) = '2015';
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

The query filters **drone ship landing failures that occurred during 2015** and extracts the month from the launch date using SUBSTR. The results show **two failed drone ship landing attempts**, both launched from **CCAFS LC-40** using **Falcon 9 v1.1 boosters**.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

SQL Query

```
SELECT
    Landing_Outcome,
    COUNT(*) AS Count_Landing
FROM SPACEXTBL
WHERE (Landing_Outcome = 'Failure (drone ship)'
       OR Landing_Outcome = 'Success (ground pad)')
       AND Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Count_Landing DESC;
```

Landing_Outcome	Count_landing
Failure (drone ship)	5
Success (ground pad)	3

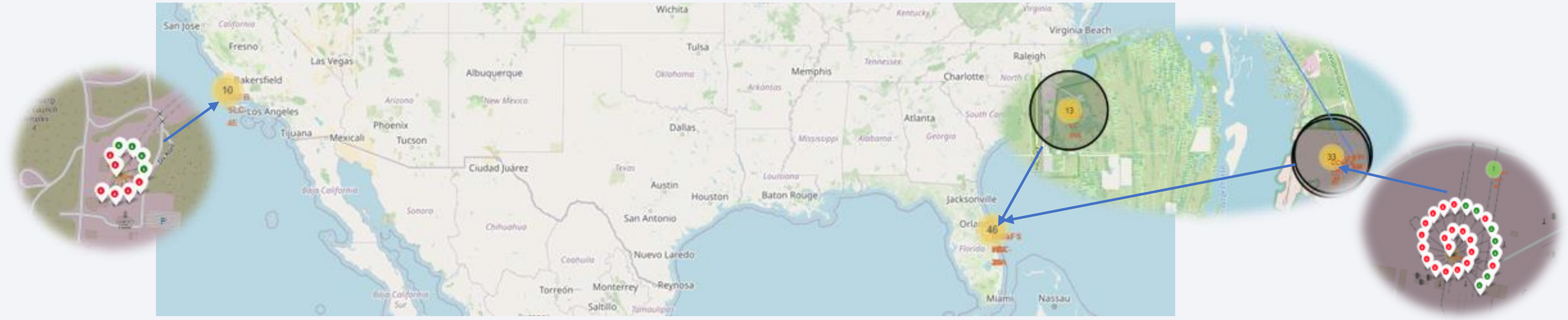
The query ranks the selected landing outcomes between **2010-06-04** and **2017-03-20** in descending order. **Failure (drone ship)** occurred **5 times**, while **Success (ground pad)** occurred **3 times**, showing that drone ship landing failures were more frequent than successful ground pad landings during the selected period.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark blue, with numerous bright yellow and orange lights representing cities and urban areas. The horizon line of the Earth is visible, separating the dark surface from the blackness of space.

Section 3

Launch Sites Proximities Analysis

Global Distribution of SpaceX Launch Sites



Key Findings

- The map displays the **geographic locations of all SpaceX launch sites** used in the dataset.
- **Markers** identify the position of each launch site on the map.
- The launch sites are concentrated in the **United States**, mainly in **Florida** and **California**.
- The map provides a clear geographic overview that supports later analyses of **launch frequency, success rates, and proximity to infrastructure**.

Launch Sites Displayed

- CCAFS LC-40
- CCAFS SLC-40
- KSC LC-39A
- VAFB SLC-4E

Launch Outcomes by Site (Success vs. Failure)

Key Findings

- The map displays **individual launch outcomes** for each SpaceX launch site.
- **Green markers** represent successful launches (Class = 1).
- **Red markers** represent failed launches (Class = 0).
- **Marker clusters** group nearby launches, making the map easier to explore and compare.
- Most launch sites show a **higher concentration of successful launches**, especially the Florida launch locations, indicating improving launch reliability over time.



VAFB SLC-4E



CCAFS LC-40



KSC LC-39A

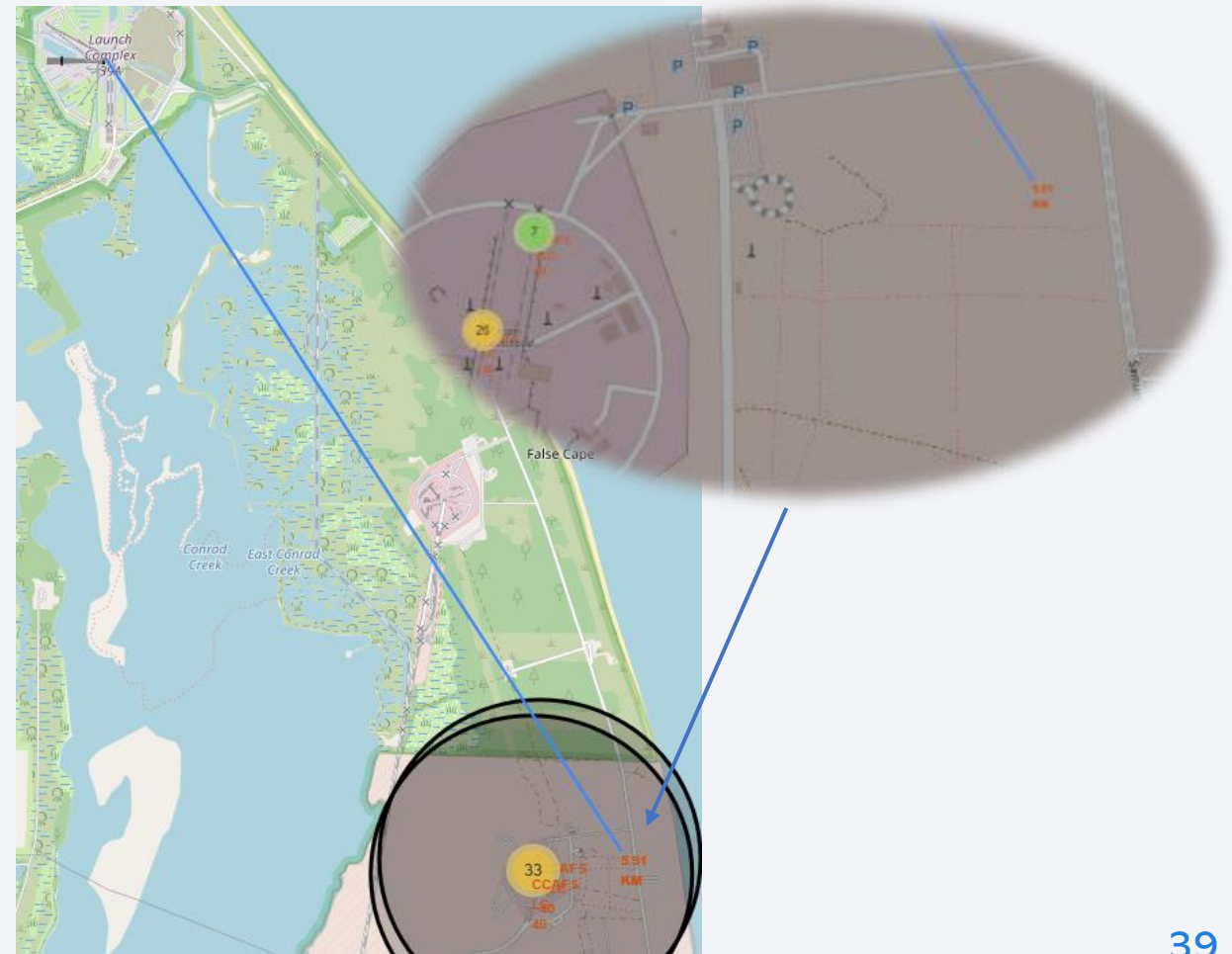


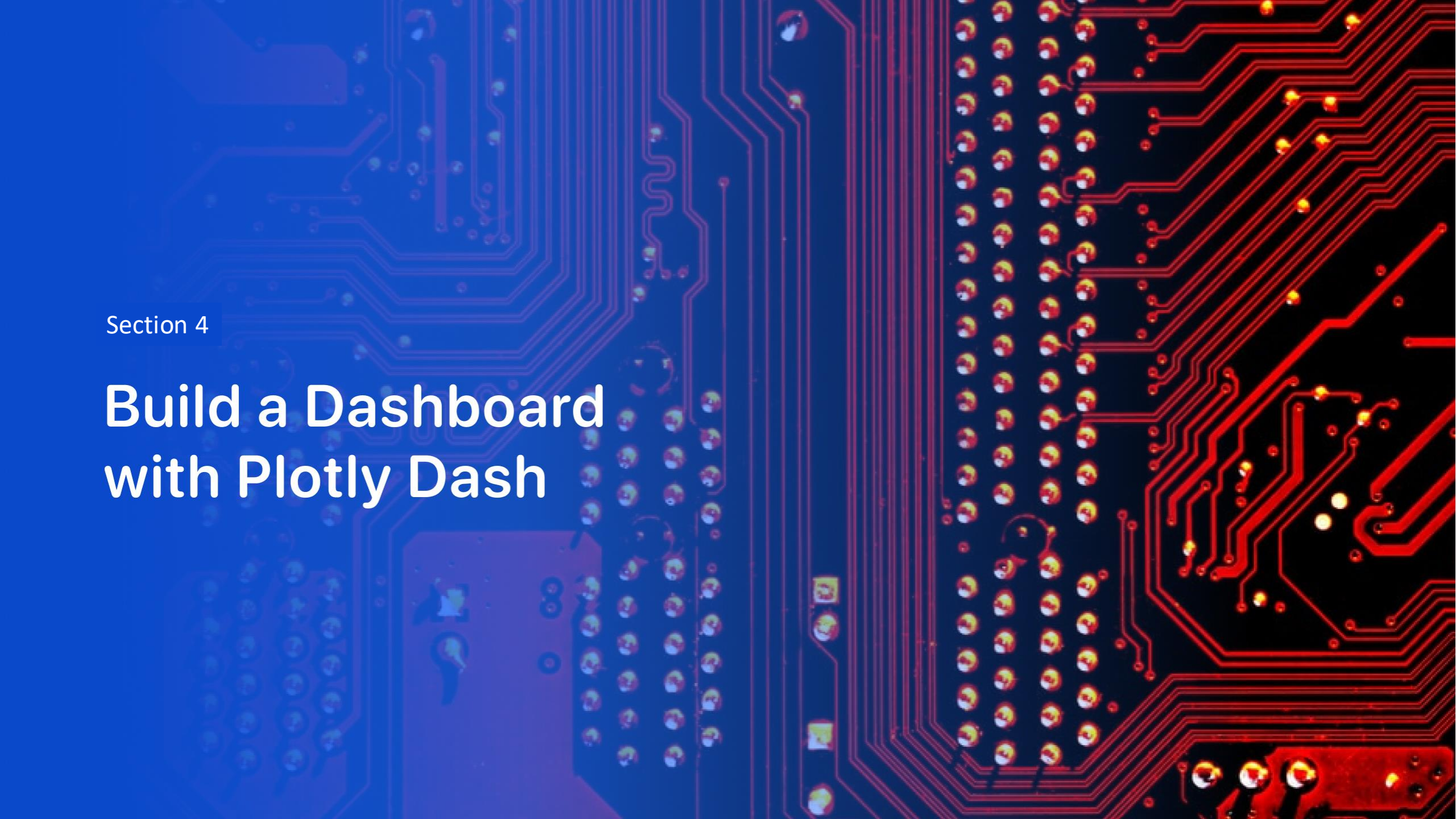
CCAFS SLC-40

Launch Site Proximity Analysis with Distance Measurement

Key Findings

- The map displays a **selected launch site** and its proximity to nearby infrastructure.
- **Distance labels** indicate the calculated distance between the launch site and Launch Complex 39A.
- The selected launch site is located **very close to the coastline**, while roads and rail infrastructure are also nearby.
- This proximity supports **transportation logistics, equipment delivery, and launch operations**, making the location well suited for space launch activities.





Section 4

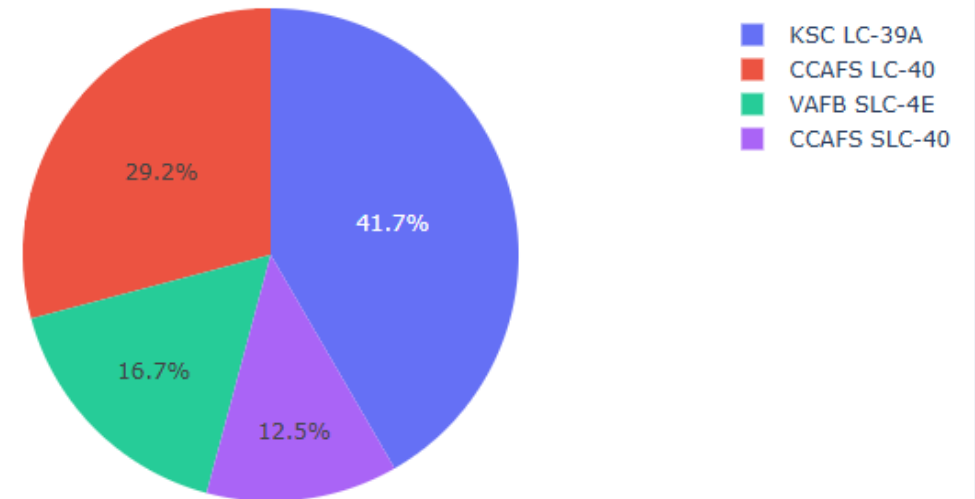
Build a Dashboard with Plotly Dash

Launch Success Count for All Launch Sites

Key Findings

- The pie chart displays the **total number of successful launches for each launch site**.
- Each slice represents one of the four SpaceX launch sites.
- The chart allows a quick comparison of **which sites contributed the most successful launches**.
- **KSC LC-39A** and **CCAFS SLC-40** account for a large share of successful missions, indicating that these sites were among the most successful and active launch locations in the dataset.

Total Success Launches by Site

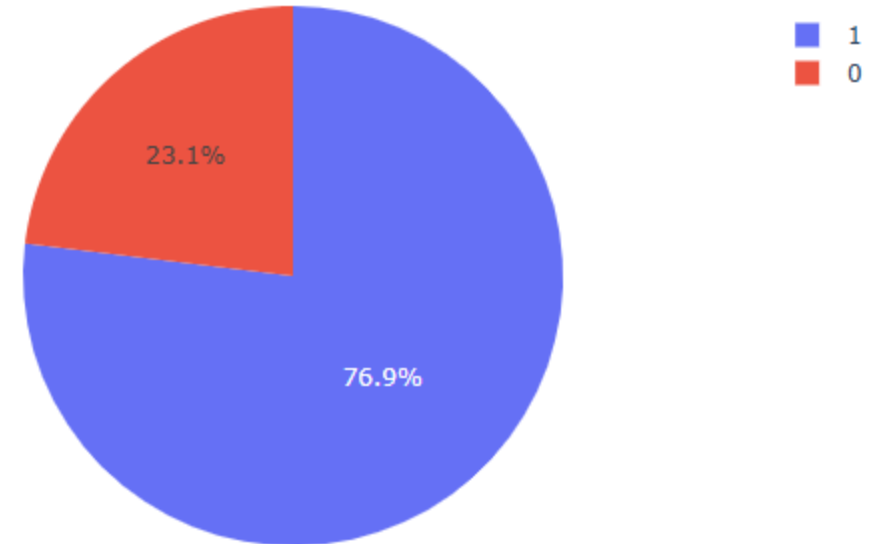


Launch Success Ratio for KSC LC-39A

Key Findings

- The pie chart displays the **success and failure distribution** for launches from **KSC LC-39A**.
- **Success (Class = 1)** occupies the largest portion of the chart.
- The chart indicates that **KSC LC-39A has the highest launch success ratio** among the launch sites in the dataset.
- This suggests that launches from KSC LC-39A achieved a **consistently high level of mission success** compared with the other launch locations.

Total Success Launches for site KSC LC-39A



Payload Mass vs. Launch Outcome Across All Sites

Key Findings

- The scatter plot shows the relationship between **payload mass** and **launch outcome** across all launch sites.
- The **x-axis** represents payload mass, while the **y-axis** represents the launch outcome (Class: 0 = Failure, 1 = Success).
- **Point colors** identify different **Booster Version Categories**.
- Lower payload ranges contain a mix of successful and failed launches, while **payloads above approximately 4,000 kg show a noticeably higher concentration of successful launches**.
- **FT, B4, and especially B5 booster versions** appear most frequently among successful launches, indicating that newer Falcon 9 boosters achieved the highest success rates across a wide range of payload masses.

Correlation between Payload and Success for all Sites





Section 5

Predictive Analysis (Classification)

Classification Accuracy

Key Findings

- The evaluated models were **Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN)**.
- The comparison shows the performance of each model on the test dataset after training and hyperparameter tuning.

Model	Cross-Validation Accuracy
Logistic Regression	0.8464
SVM	0.8482
Decision Tree	0.8893
KNN	0.8482

Conclusion

The **Decision Tree classifier** achieved the **highest classification accuracy (0.8893)** among all evaluated models, making it the best-performing predictive model for the SpaceX launch success classification task.

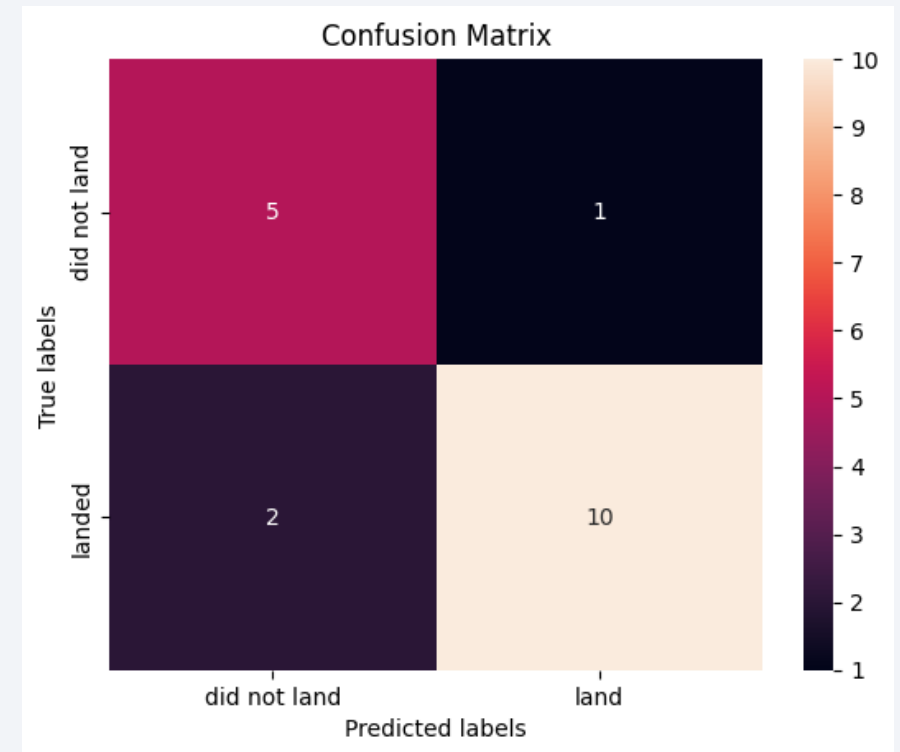
Confusion Matrix

Key Findings

- The confusion matrix evaluates the **Decision Tree classifier**, which achieved the highest overall accuracy in the predictive analysis.
- **Correct predictions:** 15 out of 18 test samples.
- **True Positives (Landed predicted as Landed):** 10
- **True Negatives (Did not land predicted as Did not land):** 5
- **False Positives:** 1
- **False Negatives:** 2

Interpretation

The model correctly identified most successful landings while producing **very few misclassifications**. The confusion matrix shows strong predictive performance and confirms that the **Decision Tree model was the most effective classifier** for predicting SpaceX booster landing outcomes in this project.



Conclusions

- SpaceX launch data was successfully collected from the **SpaceX API** and **Wikipedia web scraping**, then cleaned and prepared through a complete data wrangling process.
- Exploratory data analysis revealed that **launch site, orbit type, payload mass, and booster version** are important factors associated with launch success.
- Interactive visualizations using **Folium** and **Plotly Dash** provided geographic and operational insights, including launch site performance and payload-related success patterns.
- Several machine learning classification models were trained and evaluated, and the **Decision Tree classifier achieved the best performance** with an accuracy of approximately **0.8463**.
- The project demonstrates how **data collection, SQL analysis, visualization, interactive dashboards, and predictive modeling** can be combined to support decision-making in aerospace operations.

Appendix

Python Code Snippet

- `features_one_hot = pd.get_dummies(`
- `features,`
- `columns=['Orbit', 'LaunchSite', 'LandingPad', 'Serial']`
- `)`

SQL Query Example

- `SELECT Launch_Site, COUNT(*) AS Launches`
- `FROM SPACEXTBL`
- `GROUP BY Launch_Site;`

Additional Assets Included

- Jupyter Notebook outputs
- EDA charts
- Folium maps
- Plotly Dash dashboard screenshots
- Classification model results and confusion matrix

Appendix

GitHub Repository

All completed notebooks, Python scripts, Dash application, and project assets are available at:

<https://github.com/jhonys4444/IBM-SpaceX-Capstone>

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Repository contents:

- API Data Collection notebook
- Web Scraping notebook
- Data Wrangling notebook
- EDA Visualization notebook
- EDA SQL notebook
- Folium Map notebook
- Machine Learning notebook
- Plotly Dash application

Thank you!

